HUMBER ENGINEERING

MENG-3020
SYSTEMS MODELING & SIMULATION

LECTURE 4

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LECTURE 4
Transfer Function Approach to Modeling Dynamic Systems

- Standard Forms of System Models
- Transfer Function Model
- Block Diagrams with Transfer Function
- State-space Equations from Transfer Function

Standard Forms of System Models

- In the previous lecture, we introduced the standard forms of dynamic system models.

- The two most common standard forms for dynamic system models:
  - **State-Space Model**
    - Provides a model in time domain, and internal description of the system via state variables
    - Applicable for time-varying, nonlinear and MIMO systems

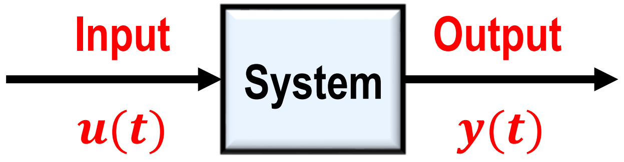
$$\begin{cases} \dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t) \end{cases}$$

  - **Transfer Function Model**
    - Provides a model in Laplace domain, and input-output description (External description)
    - Limited to linear and time-invariant (LTI) and mostly applicable to SISO or two-input two-output systems

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + \cdots + b_2 s^2 + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0}$$

Laplace Transform Properties Review

- Laplace Transform converts a time-domain function to s-domain. $\boxed{\mathcal{L}\{f(t)\} = F(s)}$
- Input-output variable of physical dynamic systems are function of time and are showing with lower case letters.
  For example, input $u(t)$ and output $y(t)$
- Laplace transform of the variables are function of s and are showing with upper case letters.
  For example, input $U(s)$ and output $Y(s)$



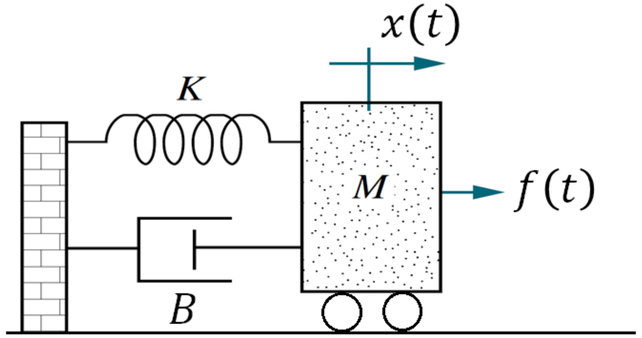
- **Linearity Property**
  - $f_1(t) \pm f_2(t) \quad \Leftrightarrow \quad F_1(s) \pm F_2(s)$
  - $kf(t) \quad \Leftrightarrow \quad kF(s)$
- For example:
  $2v(t) + 5u(t) \quad \Leftrightarrow \quad 2V(s) + 5U(s)$

- **Differentiation & Integration**
  - $\frac{df(t)}{dt}$ or $f'(t)$ or $\dot{f}(t) \quad \Leftrightarrow \quad sF(s) - f(0)$
  - $\frac{d^2f(t)}{dt^2}$ or $f''(t)$ or $\ddot{f}(t) \quad \Leftrightarrow \quad s^2F(s) - sf(0) - f'(0)$
  - $\int_0^t f(\mathrm{t})dt \quad \Leftrightarrow \quad \frac{1}{s}F(s)$
- For example:
  $2v'(t) - 3v(t) \quad \Leftrightarrow \quad 2\big(sV(s) - v(0)\big) - 3V(s)$

Input-Output Relationship in Laplace Domain

- Consider a mass-spring-damper system that is represented by the following second-order differential equation.

- Assume that the applied force $f(t)$ is the <u>input</u>, and the displacement of the mass $x(t)$ is the <u>output</u>.
$M = 50kg, B = 30\ N.s/m, K = 100\ N/m$



$$50x''(t) + 30x'(t) + 100x(t) = f(t)$$

- We can find the input-output relationship in Laplace domain by taking Laplace transform of the differential equation and assuming zero initial conditions (no initial position and no initial velocity, $x(0) = 0,\ \ x'(0) = 0$):

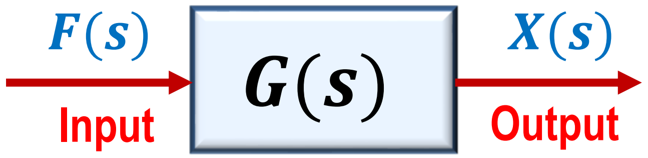
$$50\big(s^2X(s) - sx(0) - x'(0)\big) + 30\big(sX(s) - x(0)\big) + 100X(s) = F(s)$$

$$50s^2X(s) + 30sX(s) + 100X(s) = F(s) \quad \rightarrow \quad (50s^2 + 30s + 100)X(s) = F(s)$$

$$\rightarrow \quad X(s) = \underbrace{\left(\frac{1}{50s^2 + 30s + 100}\right)}_{\boldsymbol{G(s)}} F(s)$$

Transfer Function

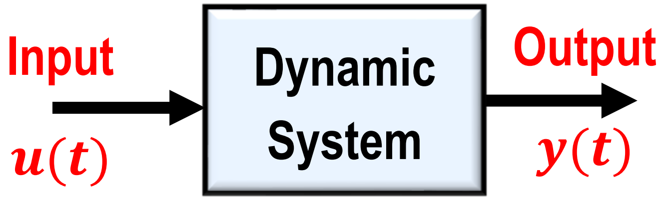
$$\boxed{\boldsymbol{X(s) = G(s)F(s)}}$$



$$\boxed{\text{Output} = G(s) \times \text{Input}}$$

Transfer Function Model

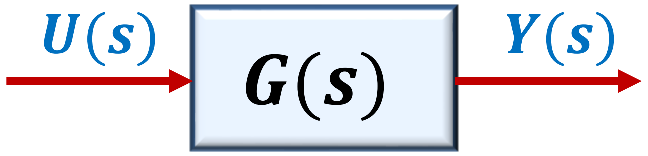
- Consider a dynamic system with input $u(t)$ and output $y(t)$
- In general, **LTI** systems can be modeled by ordinary differential equations (ODE).



- Transfer function describes the **input-output relationship** of a LTI system as a function of s in Laplace domain when **all initial conditions before applying the input are zero.**

Transfer Function $$\boxed{G(s) \triangleq \frac{Y(s)}{U(s)}}$$

- A transfer function can be represented as a block diagram with input $U(s)$ and output $Y(s)$ and the transfer function $G(s)$ as the operator in the box that converts the input to the output.



- Having a transfer function model of a system, we can find output of the system for any input and analyze performance of the system.

$$\boxed{Y(s) = G(s)U(s)}$$

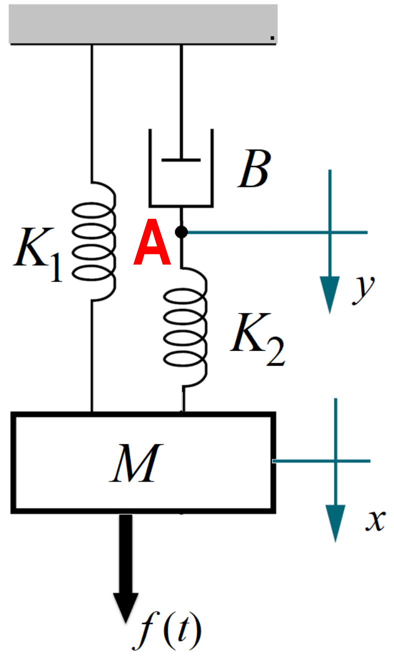
Transfer Function Model

Example 1

Consider the following mechanical system. The system is at rest initially. The displacements x and y are measured from their respective equilibrium positions.

Assume that the applied force $f(t)$ is the input, and the displacement of the mass, $x(t)$ and the displacement of massless junction A, $y(t)$ are the outputs.

Determine the transfer function models of the system. $G_1(s) = \dfrac{X(s)}{F(s)}$ $\quad G_2(s) = \dfrac{Y(s)}{F(s)}$



The equations of motion for this system are:

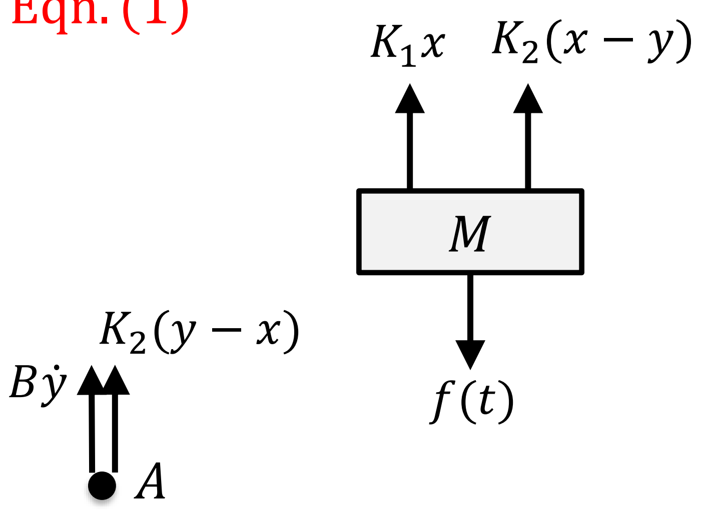
Mass M $\rightarrow$ $f(t) - K_1 x - K_2(x - y) = M\ddot{x}$

Massless Junction A $\rightarrow$ $-K_2(y - x) - B\dot{y} = 0 \rightarrow K_2(x - y) = B\dot{y}$

Take Laplace transform of these two equations, assuming zero initial conditions.

$$F(s) - K_1 X(s) - K_2\big(X(s) - Y(s)\big) = Ms^2 X(s) \rightarrow F(s) = (Ms^2 + K_1 + K_2)X(s) - K_2 Y(s) \qquad \text{Eqn. (1)}$$

$$K_2\big(X(s) - Y(s)\big) = BsY(s) \rightarrow K_2 X(s) = (Bs + K_2)Y(s) \qquad \text{Eqn. (2)}$$



To obtain the transfer function $G_1(s)$, solve Eqn. (2) for $Y(s)$ and substitute the result into Eqn. (1).

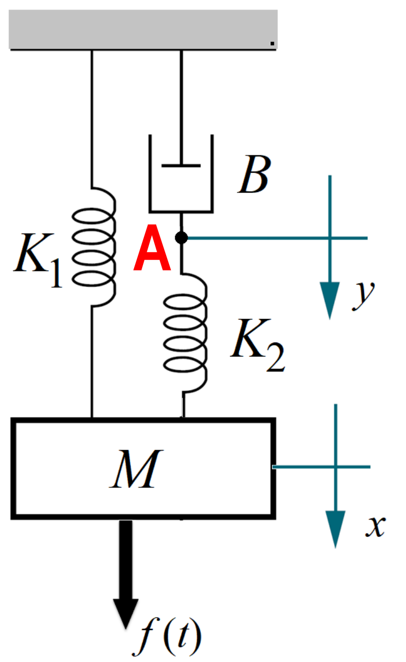
From Eqn. (2) $\rightarrow$ $Y(s) = \dfrac{K_2}{Bs + K_2} X(s)$

Substitute into Eqn. (1) $\rightarrow$ $F(s) = (Ms^2 + K_1 + K_2)X(s) - \dfrac{K_2^2}{Bs + K_2} X(s)$

Transfer Function Model

Example 1 Determine the transfer function models of the system.

$$G_1(s) = \frac{X(s)}{F(s)} \qquad G_2(s) = \frac{Y(s)}{F(s)}$$



$$F(s) = (Ms^2 + K_1 + K_2)X(s) - \frac{K_2^2}{Bs + K_2}X(s)$$

$$(Bs + K_2)F(s) = (Bs + K_2)(Ms^2 + K_1 + K_2)X(s) - K_2^2X(s)$$

$$(Bs + K_2)F(s) = \left((Bs + K_2)(Ms^2 + K_1 + K_2) - K_2^2\right)X(s)$$

$$(Bs + K_2)F(s) = (MBs^3 + K_1Bs + K_2Bs + MK_2s^2 + K_1K_2 + K_2^2 - K_2^2)\,X(s)$$

$$(Bs + K_2)F(s) = (MBs^3 + MK_2s^2 + (K_1 + K_2)Bs + K_1K_2)X(s)$$

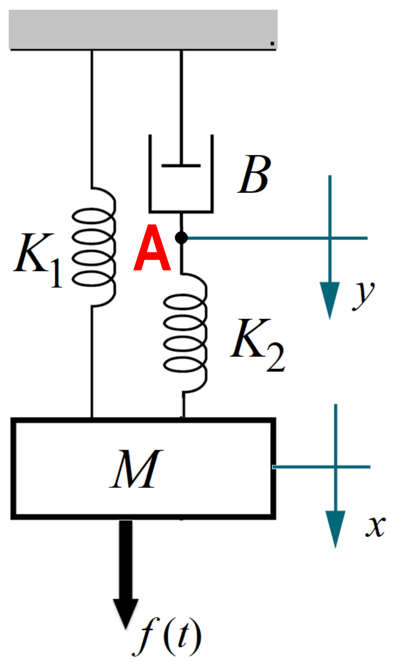
$$\boxed{\frac{X(s)}{F(s)} = \frac{Bs + K_2}{MBs^3 + MK_2s^2 + (K_1 + K_2)Bs + K_1K_2}}$$

Transfer Function $G_1(s)$

Transfer Function Model

Example 1 Determine the transfer function models of the system.

$$G_1(s) = \frac{X(s)}{F(s)} \qquad G_2(s) = \frac{Y(s)}{F(s)}$$



Recall the equations (1) and (2).

$$F(s) = (Ms^2 + K_1 + K_2)X(s) - K_2Y(s) \qquad \text{Eqn. (1)}$$

$$K_2X(s) = (Bs + K_2)Y(s) \qquad \text{Eqn. (2)}$$

To obtain the transfer function $G_2(s)$, solve Eqn. (2) for $X(s)$ and substitute into Eqn. (1)

From Eqn. (2) $\rightarrow$ $X(s) = \dfrac{Bs + K_2}{K_2}Y(s)$

Substitute into Eqn. (1) $\rightarrow$ $F(s) = \dfrac{(Ms^2 + K_1 + K_2)(Bs + K_2)}{K_2}Y(s) - K_2Y(s)$

$$K_2F(s) = (Ms^2 + K_1 + K_2)(Bs + K_2)Y(s) - K_2^2Y(s)$$

$$K_2F(s) = \left((Ms^2 + K_1 + K_2)(Bs + K_2) - K_2^2\right) Y(s)$$

$$K_2F(s) = (MBs^3 + MK_2s^2 + K_1Bs + K_1K_2 + K_2Bs + K_2^2 - K_2^2)\, Y(s)$$

$$K_2F(s) = (MBs^3 + MK_2s^2 + (K_1 + K_2)Bs + K_1K_2)\, Y(s)$$

$$\boxed{\frac{Y(s)}{F(s)} = \frac{K_2}{MBs^3 + MK_2s^2 + (K_1 + K_2)Bs + K_1K_2}}$$

Transfer Function $G_2(s)$

Transfer Function Properties

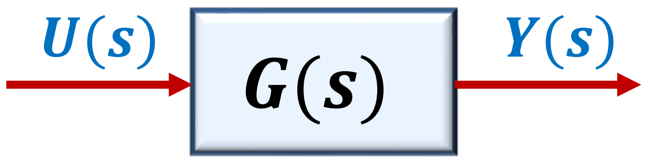
- The transfer function is equivalent to the ordinary differential equation (ODE).
  - For example, the following transfer function corresponds to the given equation:

$$\frac{X(s)}{F(s)} = \frac{0.4s + 4}{0.04s^3 + 0.4s^2 + 4s + 24} \quad \Longleftrightarrow \quad 0.04\dddot{x}(t) + 0.4\ddot{x}(t) + 4\dot{x}(t) + 24x(t) = 0.4\dot{f}(t) + 4f(t)$$

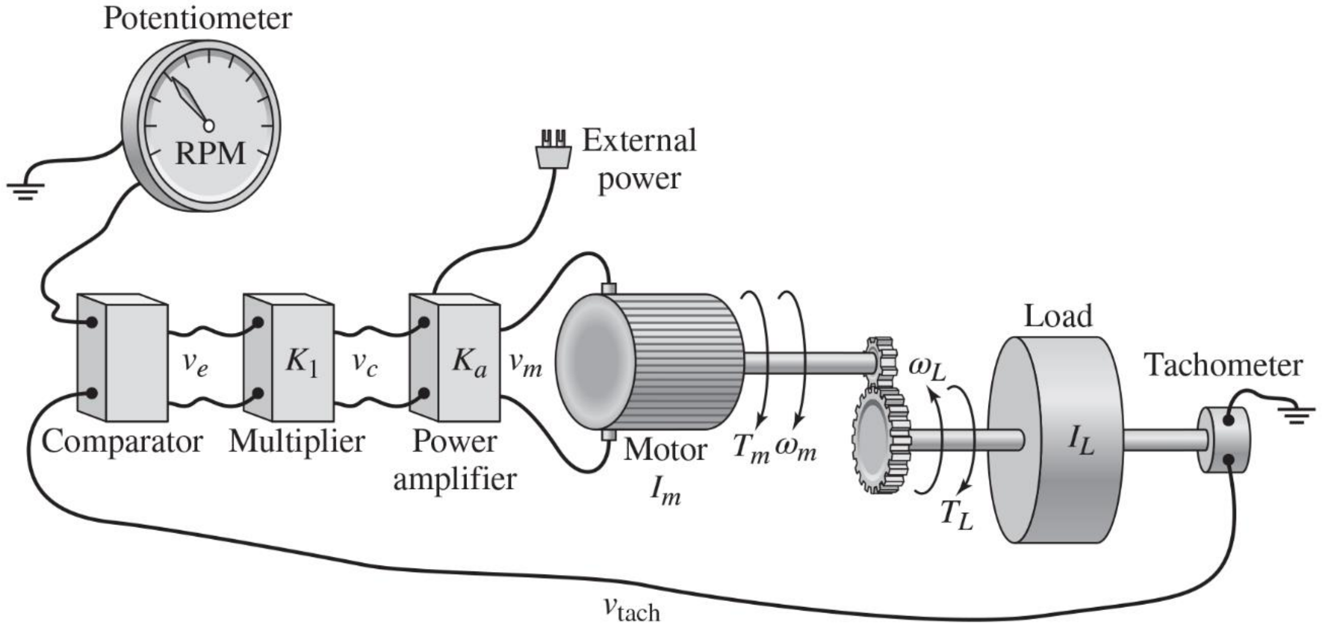
- Denominator of a transfer function is the characteristic polynomial.
- The highest power of the denominator polynomial is called the system order.
- The roots of the characteristic equation are the characteristic roots, which are also called **poles**.
- The roots of the numerator are called the **zeros**.
  - For example, in the previous transfer function:
    - The characteristic polynomial is: $0.04s^3 + 0.4s^2 + 4s + 24$
    - The system is third order.
    - The poles are: $s_{1,2} = -1.3 \pm j8.9, \quad s_3 = -7.4$
    - The zero is: $s = -10$

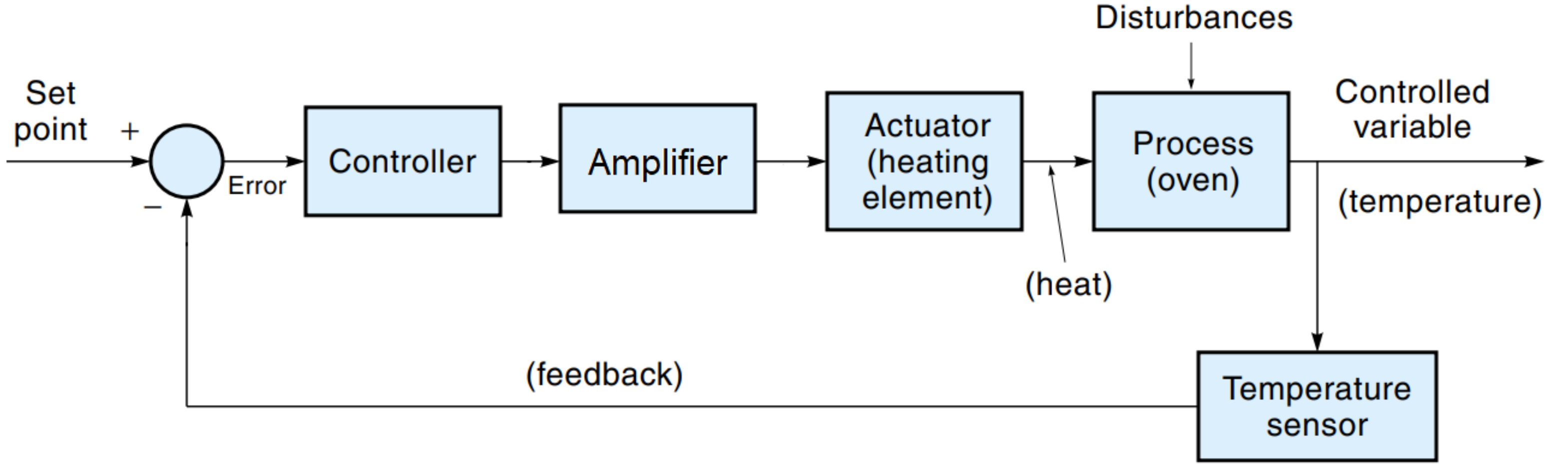
Systems Transfer Function

- The transfer function concept enables us to obtain the input-output relationship of a elements and subsystems and determine the overall transfer function model of the system.
- Therefore, we have to know how to combine the transfer functions.
- The common block diagram connections are:
  - Series or Cascade Connection
  - Parallel Connection
  - Feedback Connection



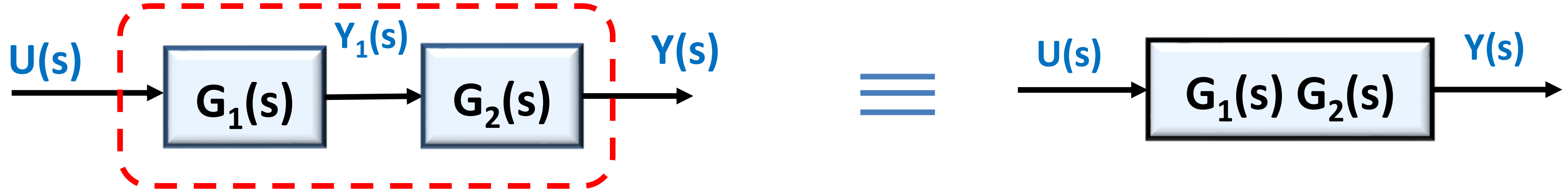
$$\boxed{Y(s) = G(s)U(s)}$$





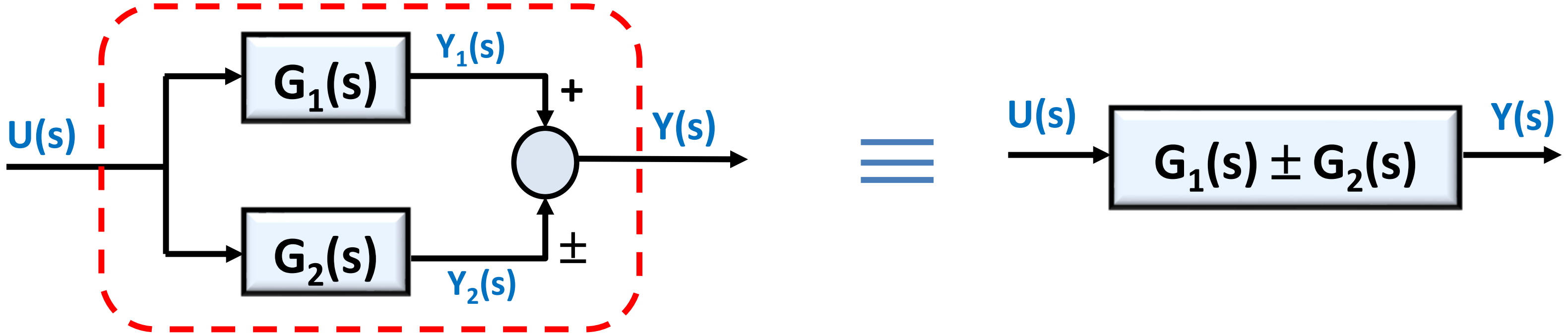
Systems Transfer Function

- **Series Connection:** The overall transfer function is equivalent to product of the individual systems transfer function.



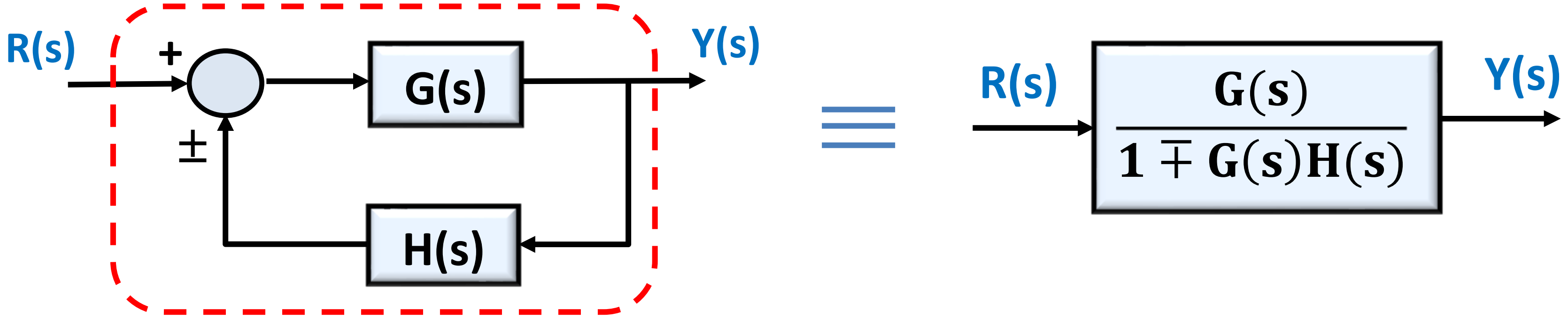
$$\frac{Y(s)}{U(s)} = G_1(s)G_2(s)$$

- **Parallel Connection:** The overall transfer function is equivalent to summation of the individual systems transfer function.



$$\frac{Y(s)}{U(s)} = G_1(s) \pm G_2(s)$$

- **Feedback Connection:** The overall transfer function is determined as below

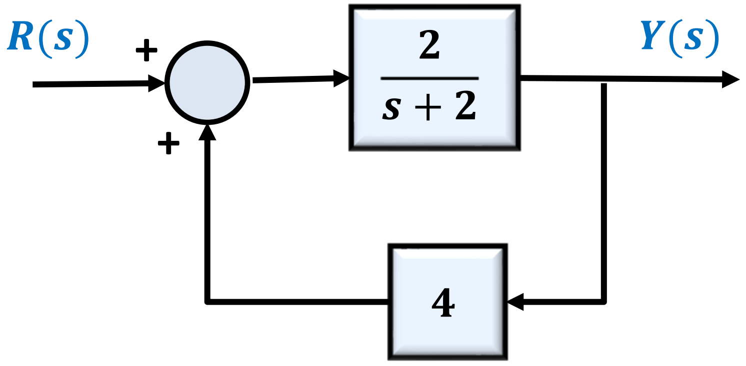


$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 \mp G(s)H(s)}$$

Systems Transfer Function

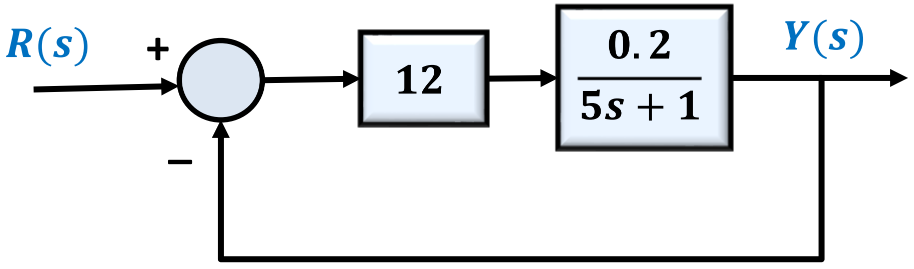
Example 2 Determine the overall transfer function from $Y(s)$ to $R(s)$ for the control systems which have shown below.

This is a single-loop with positive feedback:



$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)} = \frac{\frac{2}{s+2}}{1 - \left(\frac{2}{s+2}\right)(4)} = \frac{\frac{2}{s+2}}{1 - \frac{8}{s+2}} = \frac{\frac{2}{s+2}}{\frac{s+2-8}{s+2}} = \frac{2}{s-6}$$

This is a single-loop with negative feedback:

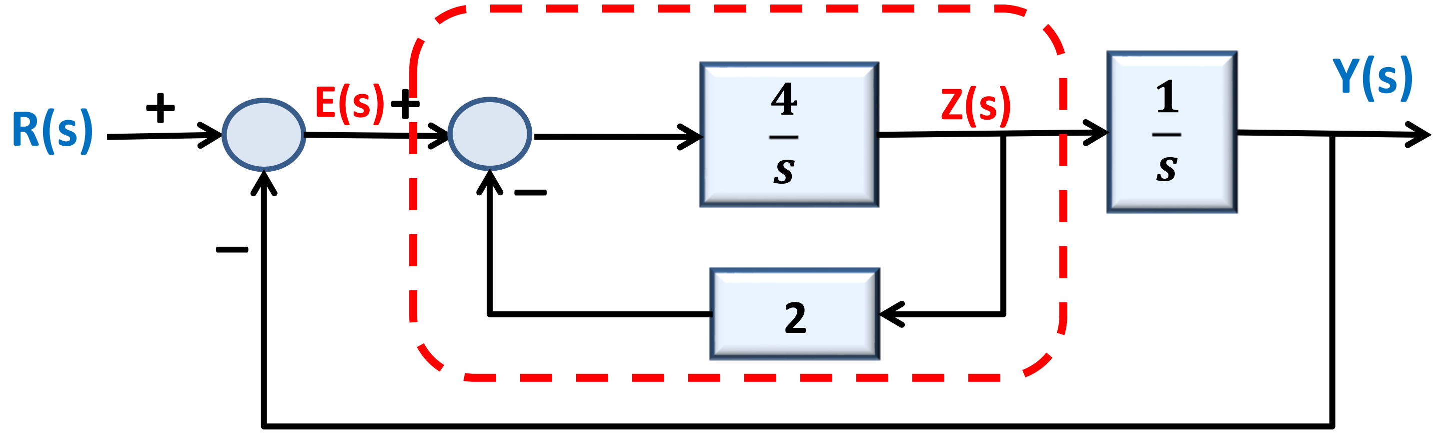


$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{(12)\left(\frac{0.2}{5s+1}\right)}{1 + (12)\left(\frac{0.2}{5s+1}\right)(1)} = \frac{\frac{2.4}{5s+1}}{1 + \frac{2.4}{5s+1}} = \frac{\frac{2.4}{5s+1}}{\frac{5s+1+2.4}{5s+1}} = \frac{2.4}{5s+3.4}$$

Systems Transfer Function

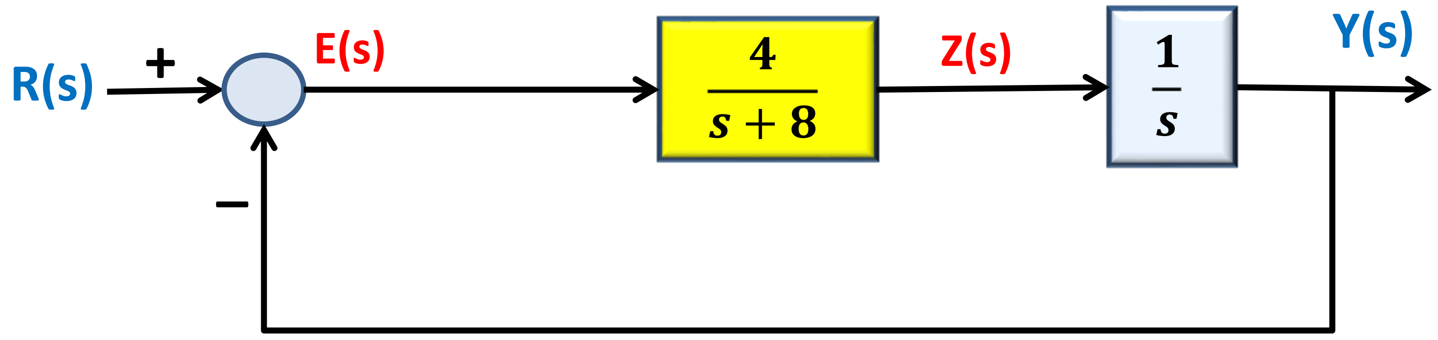
Example 3 Determine the overall transfer function $Y(s)$ to $R(s)$ for the following control system.

This is a multiple-loop system with negative feedback:



First determine the transfer function of internal feedback loop from $Z(s)$ to $E(s)$:

$$\frac{Z(s)}{E(s)} = \frac{\boldsymbol{G}}{\boldsymbol{1+GH}} = \frac{\frac{4}{s}}{1+\left(\frac{4}{s}\right)(2)} = \frac{\frac{4}{s}}{1+\frac{8}{s}} = \frac{\frac{4}{s}}{\frac{s+8}{s}} = \frac{4}{s+8}$$



Thus, the overall transfer function from $Y(s)$ to $R(s)$ is:

$$\frac{Y(s)}{R(s)} = \frac{\boldsymbol{G}}{\boldsymbol{1+GH}} = \frac{\left(\frac{4}{s+8}\right)\left(\frac{1}{s}\right)}{1+\left(\frac{4}{s+8}\right)\left(\frac{1}{s}\right)(1)} = \frac{\frac{4}{s(s+8)}}{1+\frac{4}{s(s+8)}} = \frac{\frac{4}{s(s+8)}}{1+\frac{4}{s(s+8)}} = \frac{\frac{4}{s(s+8)}}{\frac{s(s+8)+4}{s(s+8)}} = \frac{4}{s^2+8s+4}$$

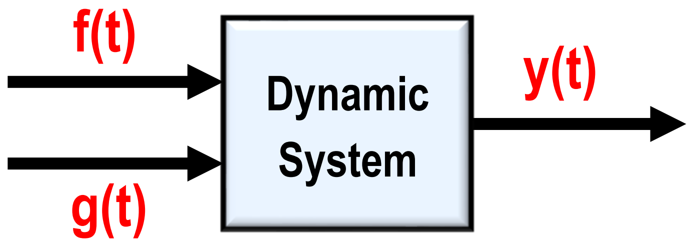
Transfer Function of MIMO Systems

- In Multi-Input-Multi-Output (MIMO) models, a particular transfer function is the ratio of the output transform over the input transform, with all the remaining inputs ignored (set to zero temporarily – superposition property).

Example 4

Obtained the transfer functions $Y(s)/F(s)$ and $Y(s)/G(s)$ for the following two-inputs one-output model of a dynamic system.

$$5\ddot{y}(t) + 30\dot{y}(t) + 40y(t) = 6f(t) + 20g(t)$$



Taking Laplace transform of both side by assuming the **zero initial conditions**

$$5s^2Y(s) + 30sY(s) + 40Y(s) = 6F(s) + 20G(s) \quad \rightarrow \quad Y(s) = \frac{6}{5s^2 + 30s + 40}F(s) + \frac{20}{5s^2 + 30s + 40}G(s)$$

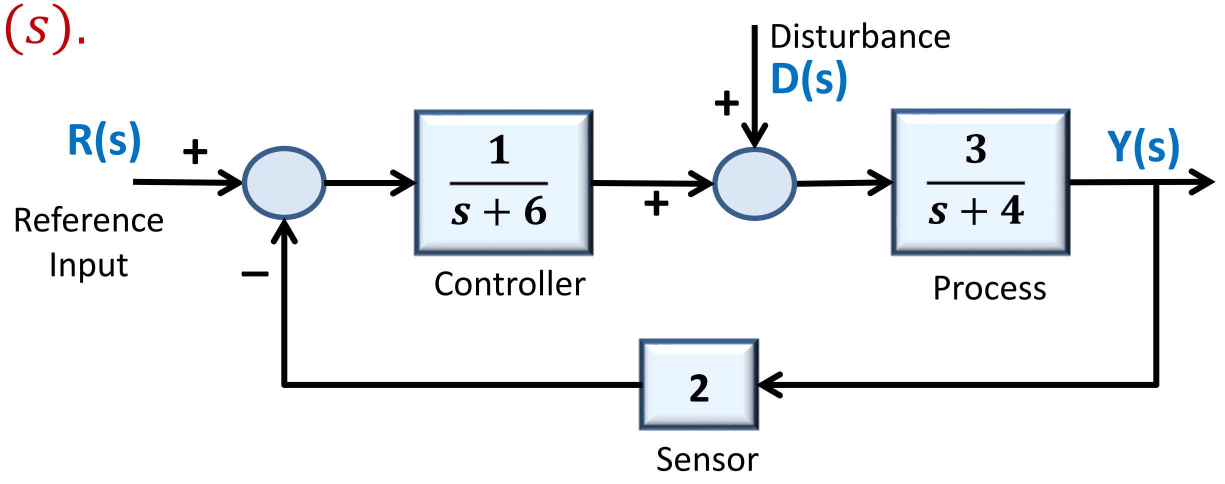
The transfer function for a specific input can be obtained by temporarily setting the other inputs equal to zero.

$$\boxed{\frac{Y(s)}{F(s)} = \frac{6}{5s^2 + 30s + 40}} \qquad \boxed{\frac{Y(s)}{G(s)} = \frac{20}{5s^2 + 30s + 40}}$$

Transfer Function Models

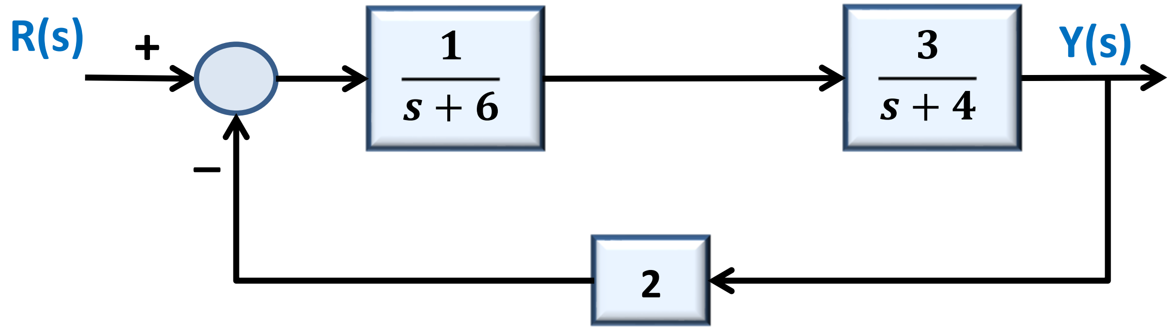
Transfer Function of MIMO Systems

Example 5 Assume the following control system with two inputs $R(s)$ and $D(s)$. Obtained the transfer functions $Y(s)/R(s)$ and $Y(s)/D(s)$.



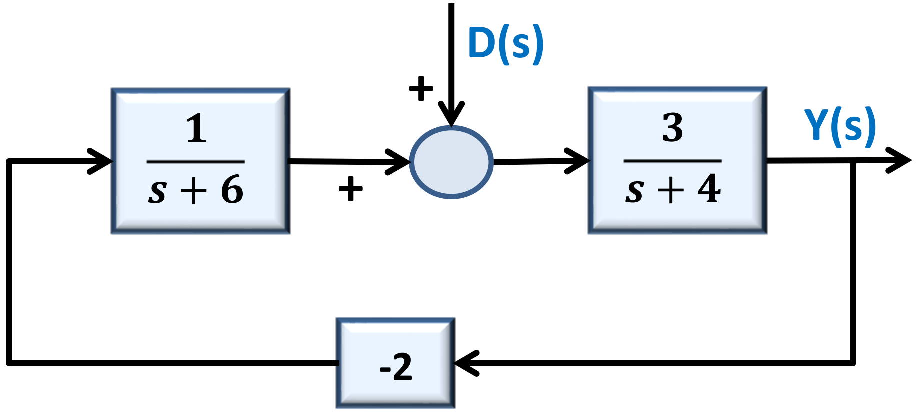
When there is more than one input to a system, the superposition principle can be used.

To obtain $Y(s)/R(s)$, set $D(s) = 0$ and redraw the diagram as shown



$$\frac{Y(s)}{R(s)} = \frac{\boldsymbol{G}}{\boldsymbol{1 + GH}} = \frac{\frac{3}{(s+6)(s+4)}}{1 + \left(\frac{3}{(s+6)(s+4)}\right)(2)} = \frac{3}{(s+6)(s+4)+6} = \frac{3}{s^2+10s+30}$$

To obtain $Y(s)/D(s)$, set $R(s) = 0$ and redraw the diagram as shown



$$\frac{Y(s)}{D(s)} = \frac{\boldsymbol{G}}{\boldsymbol{1 - GH}} = \frac{\frac{3}{s+4}}{1 - \left(\frac{3}{s+4}\right)\left(\frac{-2}{s+6}\right)} = \frac{3(s+6)}{(s+4)(s+6)+6} = \frac{3(s+6)}{s^2+10s+30}$$

The output $Y(s)$ can be written in terms of both inputs:

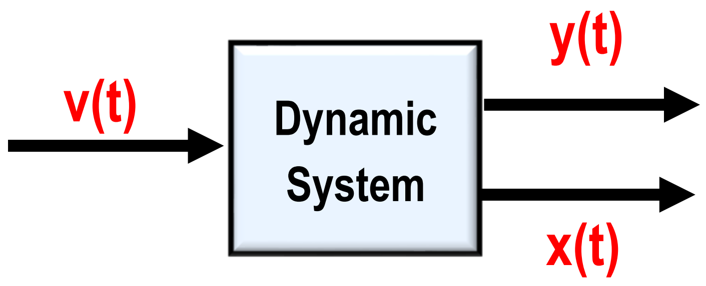
$$Y(s) = \frac{3}{s^2+10s+30}R(s) + \frac{3(s+6)}{s^2+10s+30}D(s)$$

Transfer Function of MIMO Systems

Example 6 Determine the transfer functions $X(s)/V(s)$ and $Y(s)/V(s)$ for the following system of equations:

$$\dot{x}(t) = -3x(t) + 2y(t)$$

$$\dot{y}(t) = -9y(t) - 4x(t) + 3v(t)$$



Here two outputs are specified, $x(t)$ and $y(t)$, with one input, $v(t)$.

Take Laplace transform of each equation, assuming **zero initial condition**.

$$sX(s) = -3X(s) + 2Y(s) \quad \rightarrow \quad (s+3)X(s) = 2Y(s) \qquad \text{Eqn. (1)}$$

$$sY(s) = -9Y(s) - 4X(s) + 3V(s) \quad \rightarrow \quad (s+9)Y(s) = -4X(s) + 3V(s) \qquad \text{Eqn. (2)}$$

Solve Eqn. (1) for $Y(s)$ and substitute into the Eqn. (2)

From Eqn. (1) $\rightarrow$ $Y(s) = \dfrac{s+3}{2}X(s)$

Substitute $Y(s)$ into Eqn. (2) $\rightarrow$ $(s+9)\left(\dfrac{s+3}{2}\right)X(s) = -4X(s) + 3V(s)$

Solve for $X(s)/V(s)$: $\rightarrow$ $(s+9)(s+3)X(s) = -8X(s) + 6V(s) \rightarrow (s^2 + 12s + 35)X(s) = 6V(s)$

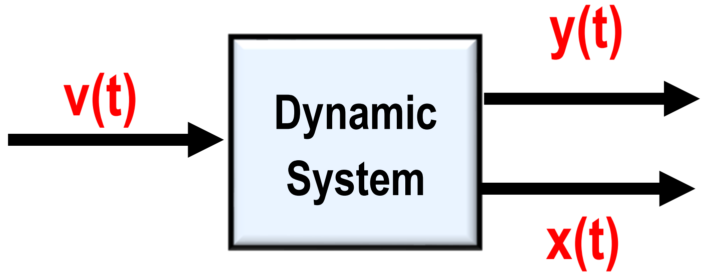
Transfer Function $\boxed{\dfrac{X(s)}{V(s)} = \dfrac{6}{s^2 + 12s + 35}}$

Transfer Function of MIMO Systems

Example 6 Determine the transfer functions $X(s)/V(s)$ and $Y(s)/V(s)$ for the following system of equations:

$$\dot{x}(t) = -3x(t) + 2y(t)$$

$$\dot{y}(t) = -9y(t) - 4x(t) + 3v(t)$$



Here two outputs are specified, $x(t)$ and $y(t)$, with one input, $v(t)$.

Take Laplace transform of each equation, assuming **zero initial condition**.

$$sX(s) = -3X(s) + 2Y(s) \quad \rightarrow \quad (s+3)X(s) = 2Y(s) \qquad \text{Eqn. (1)}$$

$$sY(s) = -9Y(s) - 4X(s) + 3V(s) \quad \rightarrow \quad (s+9)Y(s) = -4X(s) + 3V(s) \qquad \text{Eqn. (2)}$$

Solve Eqn. (1) for $X(s)$ and substitute into the Eqn. (2)

From Eqn. (1) $\rightarrow$ $X(s) = \dfrac{2}{s+3}Y(s)$

Substitute $X(s)$ into Eqn. (2) $\rightarrow$ $(s+9)Y(s) = \dfrac{-8}{s+3}Y(s) + 3V(s)$

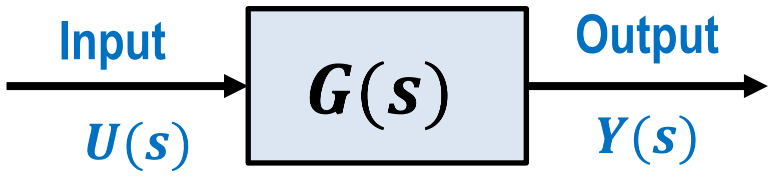
Solve for $Y(s)/V(s)$: $\rightarrow$ $(s+9)(s+3)Y(s) = -8Y(s) + 3(s+3)V(s)$ $\rightarrow$ $(s^2+12s+35)Y(s) = 3(s+3)V(s)$

Transfer Function $\boxed{\dfrac{Y(s)}{V(s)} = \dfrac{3(s+3)}{s^2+12s+35}}$

State-Space Equations From Transfer Function

- Consider an LTI, SISO system with the transfer function of G(s):

$$G(s) = \frac{\mathrm{Y}(s)}{\mathrm{U}(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$



- Determining the state space representation from the transfer function is called **realization**.
- The transfer function can be realized with minimum of *n* state variables.
- A transfer function is realizable if and only if the transfer function is proper or strictly proper.

Strictly Proper Systems ($m < n$)

$$\begin{cases} \dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) \end{cases}$$

In this case $\rightarrow$ $\mathbf{D} = \mathbf{0}$

Proper Systems ($m = n$)

$$\begin{cases} \dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t) \end{cases}$$

In this case $\rightarrow$ $\mathbf{D} = \lim_{s\to\infty} G(s)$

- General idea is deriving the differential equation from the given transfer function and then realizing the state space equations from the differential equation.

State-Space Equations From Transfer Function

Example 7 Determine the state space representation of the following transfer function.

$$\frac{Y(s)}{U(s)} = \frac{12}{s^3 + 5s^2 + 11s + 8}$$

First, find the associated differential equation

$$s^3Y(s) + 5s^2Y(s) + 11sY(s) + 8Y(s) = 12U(s) \longrightarrow \dddot{y}(t) + 5\ddot{y}(t) + 11\dot{y}(t) + 8y(t) = 12u(t)$$

$$\begin{cases} q_1(t) = y(t) \\ q_2(t) = \dot{y}(t) \\ q_3(t) = \ddot{y}(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = \dot{y}(t) \\ \dot{q}_2(t) = \ddot{y}(t) \\ \dot{q}_3(t) = \dddot{y}(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = q_3(t) \\ \dot{q}_3(t) = -8q_1(t) - 11q_2(t) - 5q_3(t) + 12u(t) \\ \text{Output} \rightarrow \quad y(t) = q_1(t) \end{cases}$$

This state variables are called Phase Variables.

$$\boxed{\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)}$$

State Equation

$$\longrightarrow \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -11 & -5 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix} u(t)$$

This is a strictly proper transfer function, $\mathbf{D} = \mathbf{0}$

$$\boxed{\mathbf{y}(t) = \mathbf{C}\boldsymbol{q}(t) + \mathbf{D}\mathbf{u}(t)}$$

Output Equation

$$\longrightarrow y(t) = [1 \quad 0 \quad 0] \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + [0]u(t)$$

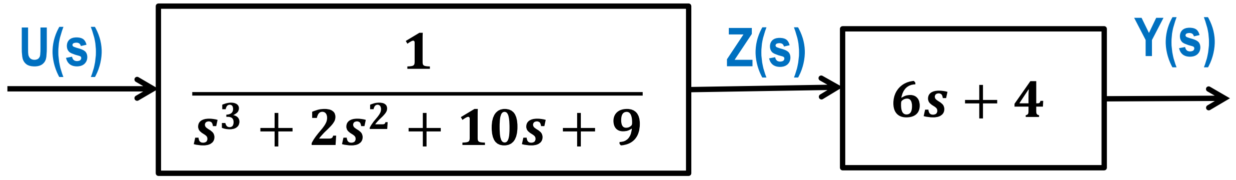
State-Space Equations From Transfer Function

Example 8 Determine the state space representation of the following transfer function.

$$\frac{Y(s)}{U(s)} = \frac{6s+4}{s^3+2s^2+10s+9}$$

Since, the numerator is a polynomial of *s*, we have to separate it into two parts as below

$$\frac{Y(s)}{U(s)} = \frac{Z(s)}{U(s)} \times \frac{Y(s)}{Z(s)} = \frac{1}{s^3+2s^2+10s+9} \times (6s+4)$$



First, find the state equation from the part with <u>denominator</u>

$$s^3Z(s) + 2s^2Z(s) + 10sZ(s) + 9Z(s) = U(s) \quad \longrightarrow \quad \dddot{z}(t) + 2\ddot{z}(t) + 10\dot{z}(t) + 9z(t) = u(t)$$

$$\begin{cases} q_1(t) = z(t) \\ q_2(t) = \dot{z}(t) \\ q_3(t) = \ddot{z}(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = \dot{z}(t) \\ \dot{q}_2(t) = \ddot{z}(t) \\ \dot{q}_3(t) = \dddot{z}(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = q_3(t) \\ \dot{q}_3(t) = -9q_1(t) - 10q_2(t) - 2q_3(t) + u(t) \end{cases}$$

Next, find the output equation by considering the effect of the <u>numerator</u>:

$$Y(s) = (6s+4)Z(s) \quad \rightarrow \quad Y(s) = 6sZ(s) + 4Z(s)$$

Take the inverse Laplace transform

$$y(t) = 6\dot{z}(t) + 4z(t) \quad \rightarrow \quad y(t) = 6q_2(t) + 4q_1(t)$$

State-Space Equations From Transfer Function

Example 8 Determine the state space representation of the following transfer function.

$$\frac{Y(s)}{U(s)} = \frac{6s + 4}{s^3 + 2s^2 + 10s + 9}$$

$$\begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = q_3(t) \\ \dot{q}_3(t) = -9q_1(t) - 10q_2(t) - 2q_3(t) + u(t) \\ y(t) = 6q_2(t) + 4q_1(t) \end{cases}$$

Therefore, the state and output equations are obtained as

$$\boxed{\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)}$$

State Equation

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -9 & -10 & -2 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$\boxed{\mathbf{y}(t) = \mathbf{C}\boldsymbol{q}(t) + \mathbf{D}\mathbf{u}(t)}$$

Output Equation

$$y(t) = \begin{bmatrix} 4 & 6 & 0 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + [0]u(t)$$

This is a strictly proper transfer function, $\mathbf{D} = \mathbf{0}$

State-Space Equations From Transfer Function

Example 9 Determine the state space representation of the following transfer function. Draw a block diagram to visualize the state variables.

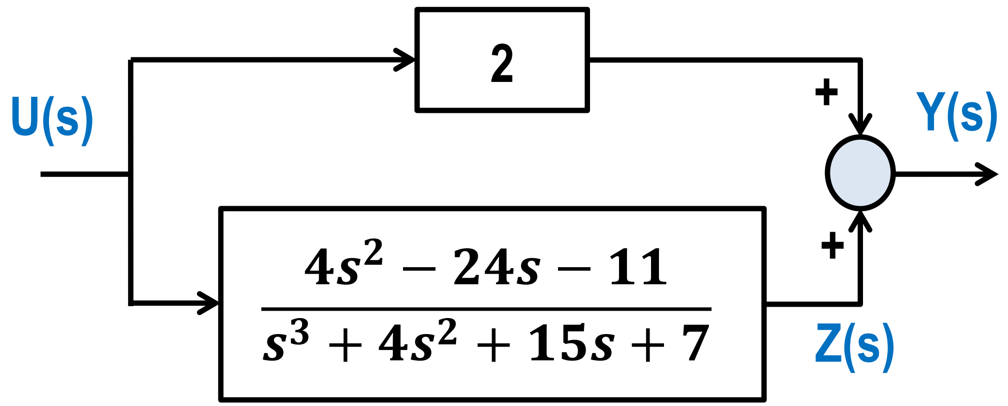
$$G(s) = \frac{Y(s)}{U(s)} = \frac{2s^3 + 12s^2 + 6s + 3}{s^3 + 4s^2 + 15s + 7}$$

$G(s)$ is a proper transfer function. First, we have to rewrite it as a summation of a constant term and a strictly proper transfer function

$$G(s) = \lim_{s\to\infty} G(s) + G_1(s) \longrightarrow G(s) = \frac{Y(s)}{U(s)} = 2 + \frac{4s^2 - 24s - 11}{s^3 + 4s^2 + 15s + 7}$$

The feed-forward matrix **D** is obtained as

$$\boxed{\mathbf{D} = \lim_{s\to\infty} G(s) = 2}$$



Determine the state space matrices, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ from the strictly proper transfer function by applying the same method in the previous example.

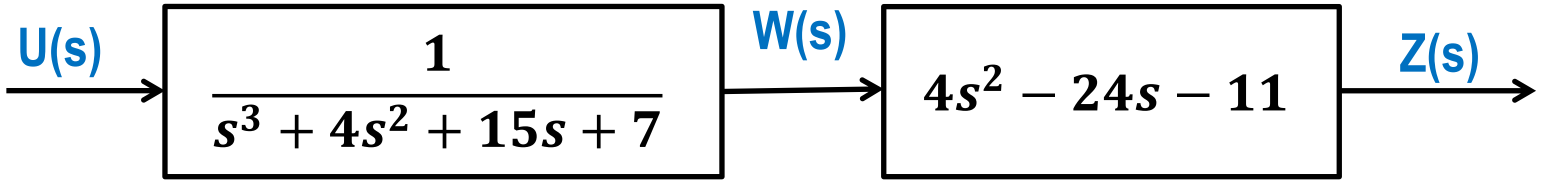
State-Space Equations From Transfer Function

Example 9 Determine the state space representation of the following transfer function. Draw a block diagram to visualize the state variables.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{2s^3 + 12s^2 + 6s + 3}{s^3 + 4s^2 + 15s + 7}$$

Determine the state space matrices, **A**, **B** and **C** from the strictly proper transfer function.

$$\frac{Z\,(s)}{U(s)} = \frac{4s^2 - 24s - 11}{s^3 + 4s^2 + 15s + 7}$$



$$\frac{Z(s)}{U(s)} = \frac{W(s)}{U(s)} \times \frac{Z(s)}{W(s)} = \frac{1}{s^3 + 4s^2 + 15s + 7} \times (4s^2 - 24s - 11)$$

Find the state equation from the part with denominator.

$$s^3W(s) + 4s^2W(s) + 15sW(s) + 7W(s) = U(s) \;\Rightarrow\; \dddot{w}(t) + 4\ddot{w}(t) + 15\dot{w}(t) + 7w(t) = u(t)$$

$$\begin{cases} q_1(t) = w(t) \\ q_2(t) = \dot{w}(t) \\ q_3(t) = \ddot{w}(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = \dot{w}(t) \\ \dot{q}_2(t) = \ddot{w}(t) \\ \dot{q}_3(t) = \dddot{w}(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = q_3(t) \\ \dot{q}_3(t) = -7q_1(t) - 15q_2(t) - 4q_3(t) + u(t) \end{cases}$$

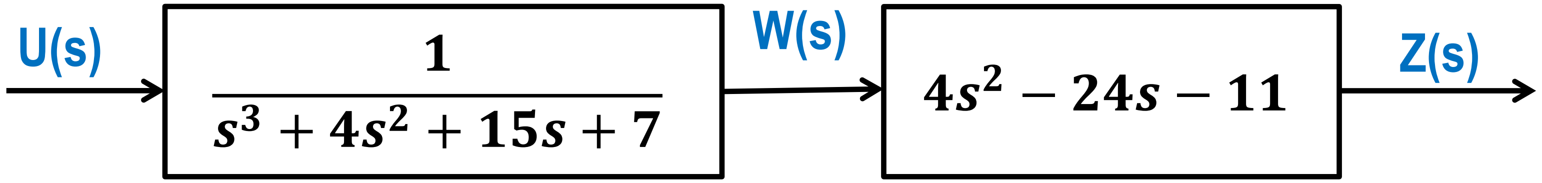
State-Space Equations From Transfer Function

Example 9 Determine the state space representation of the following transfer function. Draw a block diagram to visualize the state variables.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{2s^3 + 12s^2 + 6s + 3}{s^3 + 4s^2 + 15s + 7}$$

Determine the state space matrices, **A, B** and **C** from the strictly proper transfer function.

$$\frac{Z(s)}{U(s)} = \frac{4s^2 - 24s - 11}{s^3 + 4s^2 + 15s + 7}$$



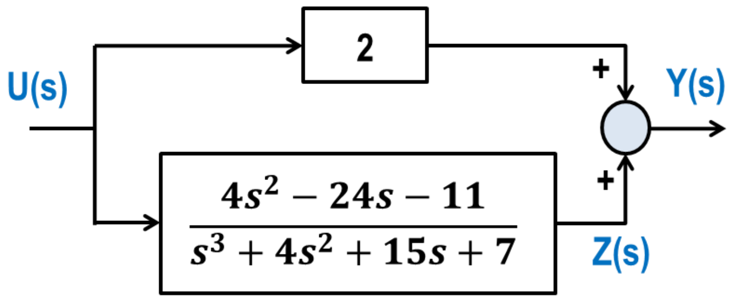
$$\frac{Z(s)}{U(s)} = \frac{W(s)}{U(s)} \times \frac{Z(s)}{W(s)} = \frac{1}{s^3 + 4s^2 + 15s + 7} \times (4s^2 - 24s - 11)$$

Next, find the output equation by considering the effect of the block with the numerator

$$Z(s) = (4s^2 - 24s - 11)W(s) \quad \rightarrow \quad Z(s) = 4s^2W(s) - 24sW(s) - 11W(s)$$

Take the inverse Laplace transform

$$z(t) = 4\ddot{w}(t) - 24\dot{w}(t) - 11w(t) \quad \rightarrow \quad z(t) = 4q_3(t) - 24q_2(t) - 11q_1(t)$$



The output is $\rightarrow$ $y(t) = z(t) + 2u(t) \quad \rightarrow \quad y(t) = 4q_3(t) - 24q_2(t) - 11q_1(t) + 2u(t)$

State-Space Equations From Transfer Function

Example 9 Determine the state space representation of the following transfer function. Draw a block diagram to visualize the state variables.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{2s^3 + 12s^2 + 6s + 3}{s^3 + 4s^2 + 15s + 7}$$

$$\begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = q_3(t) \\ \dot{q}_3(t) = -7x_1(t) - 15q_2(t) - 4q_3(t) + u(t) \\ y(t) = 4q_3(t) - 24q_2(t) - 11q_1(t) + 2\mathrm{u}(\mathrm{t}) \end{cases}$$

$$G(s) = \frac{Y(s)}{U(s)} = 2 + \frac{4s^2 - 24s - 11}{s^3 + 4s^2 + 15s + 7}$$

Therefore, the state and output equations are obtained as

$$\boxed{\dot{\mathbf{q}}(t) = \mathbf{Aq}(t) + \mathbf{Bu}(t)}$$

State Equation

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -7 & -15 & -4 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$\boxed{\mathbf{y}(t) = \mathbf{C}\boldsymbol{q}(t) + \mathbf{Du}(t)}$$

Output Equation

$$y(t) = \begin{bmatrix} -11 & -24 & 4 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix} + [2]u(t)$$

G(s) is a proper transfer function, $\mathbf{D} \neq \mathbf{0}$

THANK YOU

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